AI-Powered Fitness App

Project report

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Project type: Flutter mobile application

Overview

This project is a Flutter-based fitness application designed to help users monitor nutrition, hydration, activity, and progress over time. The app combines Firebase-based authentication and data storage with AI-assisted features for food recognition, body analysis, and health insights.

Project scope and purpose

The application aims to support a healthier lifestyle by letting users log food, track calorie and protein intake, view daily totals, monitor water intake, review lifestyle insights, and keep personal progress history.

Technical stack

Frontend: Flutter + Dart; state management: Provider; backend: Firebase Authentication, Cloud Firestore, Firebase Storage; APIs: Gemini for AI analysis and Cloudinary for image uploads; utilities: intl, image_picker, fl_chart, shared_preferences, google_fonts, sensors_plus, permission_handler.

Key modules

The app includes login/register flows, user profiles, activity screens, nutrition tracking, insights, and a dedicated body analysis module. The main application entry is in lib/main.dart, with key features split across screens, providers, and services.

Food service focus

The file lib/services/food_service.dart is a central service layer for user nutrition logic. It handles Cloudinary uploads, Gemini image analysis, Firestore storage, daily total calculation, and food entry deletion. The main methods are analyzeImage(), addFoodEntry(), getTodaysFoodEntries(), getDailyTotals(), and deleteFoodEntry().

Setup notes

Before deployment, the project requires valid Firebase configuration, real Gemini API credentials, and Cloudinary settings. Several placeholders are present in the code and must be replaced before production use.

Conclusion

This project is a modular, extensible fitness app with strong AI and cloud integration. Its main value is the combination of health tracking and automated nutrition analysis, while production readiness depends on proper credential configuration and service validation.

File reviewed: lib/services/food_service.dart

This service manages user-scoped food entries in Firestore and supports AI-powered image-to-nutrition analysis. It currently leverages Firebase Auth to get the current user, stores entries for each user in collection users/{uid}/food_entries, and computes daily totals from nutrition_history records.

- Food image upload to Cloudinary
- Gemini API nutritional analysis using the uploaded image
- Firestore storage of food entries with timestamp and image URL
- Daily retrieval of food history and nutrition totals
- Support for deleting records and handling auth-dependent state